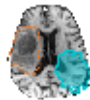


(a) Artificial training-hole examples

Provided (fixed)



1.3% of brain
Mixture: blob



8.1% of brain

Tumor-only (random)



4.2% of brain
Mixture: ellipsoid



14.6% of brain

Mixture: tumor



4.8% of brain

 Artificial hole
+ supervised target
 Real tumor
(excluded)

(b) Artificial-hole size

thin: 10--90% thick: middle 50% circle: median

Provided
 $n = 100$

Tumor-only
 $n = 500$

Mixture
 $n = 500$

